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The intrinsic mechanism of corrosion resistance for FCC high entropy alloys

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The concept of high entropy alloys (HEAs) or multi-principal component alloys has inspired a great progress of physical metallurgy along with several unusual phenomena. The high corrosion resistance of HEAs was frequently mentioned but without convincing explanations. In this paper, the intrinsic mechanism of corrosion resistance in FCC HEAs was revealed by designing equal atomic alloys with single solid solution phase. The results showed that the Cr element in FCC HEAs played the dominant role in the corrosion resistance rather than the simple structure from high entropy effect or uniform element distribution.

high entropy alloys, corrosion resistance, chromium

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1 Introduction

Conventional alloys are based on single principal component, such as Fe for steel, Mg for magnesium alloys. Different from conventional alloys, multi-principal component alloys or high entropy alloys (HEAs) consist of several principal components and without a base of single component [1–8]. The idea of HEAs greatly extends the area of alloys. Up to now, more than 300 HEAs involving around 30 elements have been designed, forming a novel family of exciting metallic materials. Because of the high mixing configuration entropy, HEAs tend to form solid solution phases with simple crystal structure. Based on this fact, HEAs have been considered to be corrosion resistant [9].

The corrosion resistance of dozens of HEAs has been investigated. Chen et al. [10,11] compared the corrosion behavior of $Cu_{0.5}NiAlCoCrFeSi$ with 304 stainless steel (304SS), and the results showed that $Cu_{0.5}NiAlCoCrFeSi$ was better than 304SS on the general corrosion in NaCl and H_2SO_4 aqueous solutions but worse than 304SS on the pitting corrosion in chloride environment. Some experiments also confirmed the effects of typical elements on the corrosion resistance of HEAs. Addition of Cu into FeCoCrNi was detrimental to the corrosion resistance because of the formation of interdendritic Cu-rich phase [12]. In the $Al_{0.5}CoCrFeNi$, the addition of Al into the CoCrFeNi alloy led the formation of (Al, Ni)-rich BCC phase and caused galvanic corrosion in NaCl solution [13]. Addition of Mo into the $Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_x$ alloys was helpful to the corrosion resistance in NaCl aqueous solution [14], while detrimental in H_2SO_4 aqueous solution, because the (Cr, Mo)-rich σ phase reduced the concentration of Cr in the matrix phase [15]. The addition of Nb to CoCrCuFeNi was reported to decrease the corrosion current density by forming stable surface oxide films [16]. All these previous investigations indicated that the corrosion behaviors of HEAs depended on the electrolytes, compositions and microstructures. The high corrosion resistance of HEAs is usually attributed to the simple phase composition and uniform

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element distribution [17,18]. The former factor could prevent the galvanic corrosion and the latter could prevent the selective corrosion. However, previous investigations showed that some HEAs were corrosion resistant while some were not [12,13,16]. It seems that the intrinsic reason of high corrosion resistance in HEAs is still uncovered.

In this paper, we tried to find out the relationships between the characteristics of HEAs and their corrosion resistance. In HEAs, the high configuration entropy, simple phase structure and the uniform element distribution are the main features with contribution to the corrosion resistance. Most importantly, the high entropy effect is the endorsement of the high entropy alloys. Therefore, we designed a series of alloys, pure Ni, equiatomic NiCo, NiCoFe, NiCoCr and NiCoCrFe to reveal the contribution of entropy and simple structure on the corrosion resistance. All these alloys are single solid solution [19,20] in the as-cast state along with simple phase structure, while the mixing configurational entropy increases gradually from the single metal to the quaternary alloy. The equiatomic design is essential in HEAs and the NiCoCrFe has been widely investigated as a base of HEAs [21]. The intrinsic mechanism of corrosion behaviors in HEAs will be revealed based on this careful design.

2 Experiment

2.1 Materials preparation

Elements nickel (Ni), cobalt (Co), iron (Fe), and chromium (Cr) with purity of 99.95 wt.% were used as raw materials. Alloys were prepared by using electric arc furnace with electromagnetic stirring for five times, and then solidifying in a water-cooled copper hearth. No obvious change in the mass of the samples was detected after solidification. The cast ingots were 60 mm in length and 10 mm in diameter. Then the samples were cut into pieces of 2 mm in thick and 10 mm in diameter. All the specimens were mechanically grounded using a series of 240#, 800#, 1500# silicon carbide (SiC) grit papers in sequence, then cleaned with ultrasound and ultrapure water. The electrochemical results are reproducible after these processing processes and the curves were measured at least three times to ensure the reproducibility. Finally, the as-cast specimens were cold mounted with the flat specimen holder kit to give an exposed surface area of 0.5 cm^2. All the curves are normalized during the data processing. The current density is the parameter used in the electrochemical measurements and the exposed area is constant in all the text.

2.2 Electrochemical measurements and surface characterization techniques

The electrochemical properties of pure Ni, and equal-atomic NiCo, NiCoCr, NiCoFe, NiCoCrFe alloys were tested in 0.1 mol/L (M) HCl and 0.1 M NaCl aqueous solutions respectively with the same chloride ion concentration but different pH at room temperature. The potentiodynamic polarization curves were measured with a typical three-electrode cell. The cell consists of a specimen as the working electrode (WE), a platinum (Pt) plate (2 cm × 2 cm) as a counter electrode (CE) and a reference electrode (RE). The RE is a saturated calomel (sat'd KCl) reference electrode (SCE) in 0.1 M NaCl solutions. In the environment of 0.1 M HCl, the RE is a silver/silver chloride (Ag/AgCl) with 3.5 M KCl. The distance between WE and CE is 20 mm; the distance between WE and RE, and the distance between RE and CE are both 24 mm.

The electrochemistry tests were performed with electrochemical work station of Princeton Applied Research (PARSTAT 4000). Cyclic polarization curves were obtained in 0.1 M NaCl with a scan rate of 1 mV/s from an initial potential −0.8 V versus the open current potential (OCP) to a final potential 1.4 V_{RE}, where the potential reversed and returned to where the polarization began. The potentiodynamic polarization measurement in 0.1 M HCl solution started at −0.8 V_{OCP} with scanning rate of 1 mV/s to 1.4 V_{RE}. In each measurement, the specimen was immersed in the prepared solutions at first and the OCP was recorded more than 30 min to yield a steady-state potential with a fluctuation less than 2 mV in 10 min. After the electrochemical measurements, the corroded surfaces were analyzed by using scanning electron microscopy (SEM), equipped with energy disperse spectroscopy (EDS), as well as X-ray photoelectron spectroscopy (XPS). XPS analyses were carried out with a Kratos Axis Ultra DLD employing monochromatic Al K_α (1486.6 eV) radiation, and the XPS spectra were corrected for charging by taking the C 1s spectrum for adventitious C to be at a binding energy of 284.8 eV. Survey spectra were recorded on the samples of NiCoCr and NiCoCrFe corroded in 0.1 M NaCl, followed by high resolution XPS spectra for Ni 2p, Co 2p, Cr 2p, Fe 2p, C 1s, and O 1s regions. The XPS spectra were analyzed using the Casa XPS software.

3 Results and analyses

The cyclic polarization curves in 0.1 M NaCl solution for different alloys are shown in Figure 1. The solid arrows close to the forward and reversed branches indicate the potential scan direction. Table 1 shows the corresponding electrochemical parameters. There are significant differences on the corrosion resistance among the alloys. The corrosion current density (i_{corr}) was obtained from the intersection of corrosion potential (E_{corr}) and the tangent in cathodic Tafel region. Furthermore, in order to investigate the pitting corrosion behavior, we also confirmed the protection potential (E_{prot}), where the forward and reversed scan curves cross in the passive region. The positive hysteresis, i.e., reversed scan current density exceeds the forward current density observed in Figure 1, indicates the appearance of pitting corrosion and the area

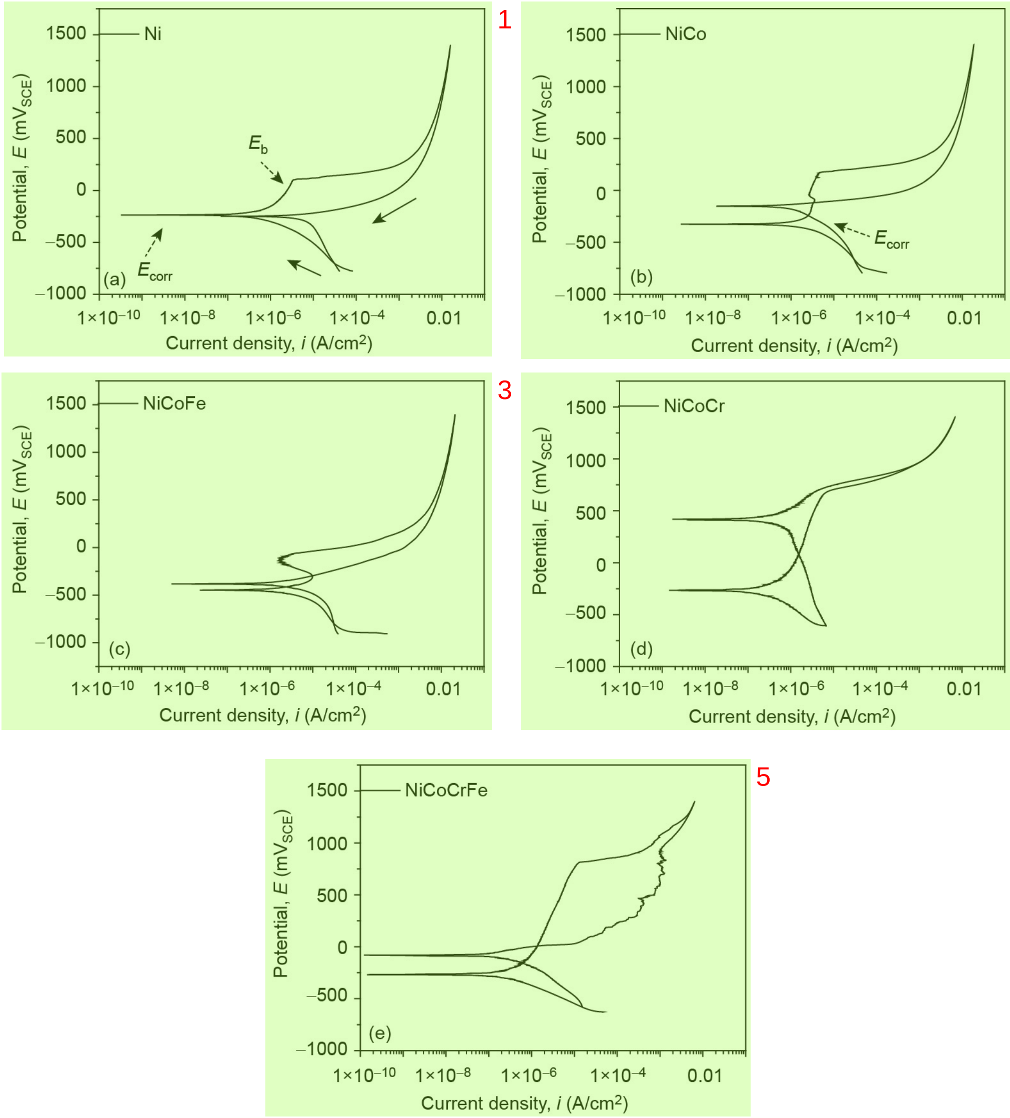

Figure 1 Cyclic polarization curves of alloys in 0.1 M NaCl. (a) Ni; (b) NiCo; (c) NiCoFe; (d) NiCoCr; (e) NiCoCrFe.

Table 1 Electrochemical parameters of the alloys in 0.1 M NaCl

Designation	E_{corr} (mV_{RE})	i_{corr} (nA/cm^2)	E_{pp} (mV_{RE})	E_b (mV_{RE})	E_{prot} (mV_{RE})	ΔE (mV)
Ni	−236	80	−98	103	–	201
NiCo	−326	290	−253	146	−140	399
NiCoFe	−448	440	−334	−143	−323	191
NiCoCr	−266	28	−139	655	–	794
NiCoCrFe	−269	40	−177	805	−17	982

contained within the loop between reversed and forward scan curves has to do with the amount of pit propagation that occurs during the cycle [22]. The established pits will stop growing and new pits will not initiate below the potential E_{prot}. However, new pits will occur over the potential E_b. In the potential range between E_{prot} and E_b, new pits will not initiate, but old ones will grow up. The lowest i_{corr} of NiCoCr means that NiCoCr possesses the lowest corrosion rate at E_{corr}. Compared with pure Ni, the passive region width (ΔE) of NiCo was much wider, indicating a beneficial effect of Co. ΔE was got from the difference of primary passive potential (E_{pp}) and E_b. For NiCoFe, it did not show distinct passive region but the active-passive region appeared. Besides, the differences among Ni, NiCo and NiCoFe enlarge in transpassive region (the region where potential is beyond E_b). As Cr element was added into the alloys, the ΔE was extremely extended and the value of i_{corr} decreased for alloy NiCoCr and NiCoCrFe. When Fe element was added into alloy NiCoCr, the ΔE became wider but i_{corr} increased. What's more, comparing the electrochemical behaviors of NiCoCr and NiCoCrFe, we can notice that the area of NiCoCrFe within the loop between reversed and forward scan curves is much bigger than that of NiCoCr, indicating that NiCoCrFe was corroded more easily with pitting corrosion. Therefore, we can conclude that NiCoCr showed better corrosion resistance.

Figure 2 shows the SEM images of the corroded alloys surfaces in 0.1 M NaCl. There are many pits on the surface of pure Ni, NiCo and NiCoFe. NiCoFe shows the worst corrosion resistance along with much more pits. However, there is no obvious pit on the corroded surfaces of NiCoCr and NiCoCrFe, even in the magnification pictures the surfaces are smooth only with a few of corrosion pits. The results of corrosion resistance from SEM images are consistent with that from the potentiodynamic polarization curves.

Figure 3 shows the potentiodynamic polarization curves of alloys in 0.1 M HCl. Table 2 is the corresponding electrochemical parameters. For pure Ni, NiCo, NiCoFe, the region of passivation is not shown. It is clear that the corrosion resistance of the alloys in 0.1 M HCl is worse than that in 0.1 M NaCl with much higher current density, which is due to the decisive effect of pH on corrosion. Here, the NiCoCr and NiCoCrFe are still far better than the other alloys. NiCoCr possesses best corrosion resistance with smallest current density in passive region and widest passivation region ΔE. For pure Ni, NiCo and NiCoFe, there is no passivation and the similar curves indicate that the oxide/hydroxide of Ni, Co and Fe are unstable and dissolved in 0.1 M HCl. For NiCoCr alloy, there are multiple passivations at the potential close to E_b in passive region, where the passivation and breakdown compete with each other and corrosion pits occur. In the passive region of NiCoCrFe, the current density decreased when the passivation began and the current density increased when

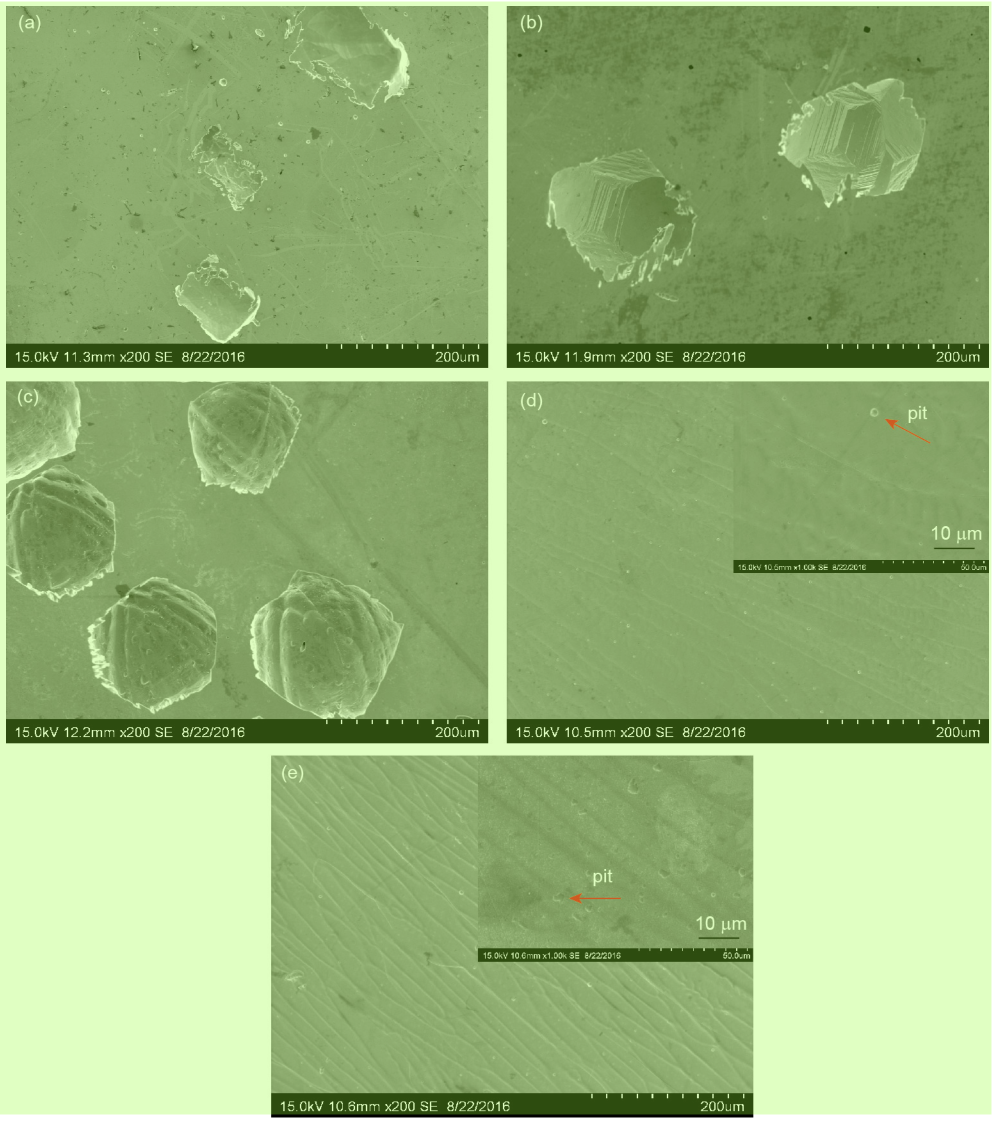

Figure 2 (Color online) The corroded surfaces of alloys in 0.1 M NaCl. (a) Ni; (b) NiCo; (c) NiCoFe; (d) NiCoCr; (e) NiCoCrFe.

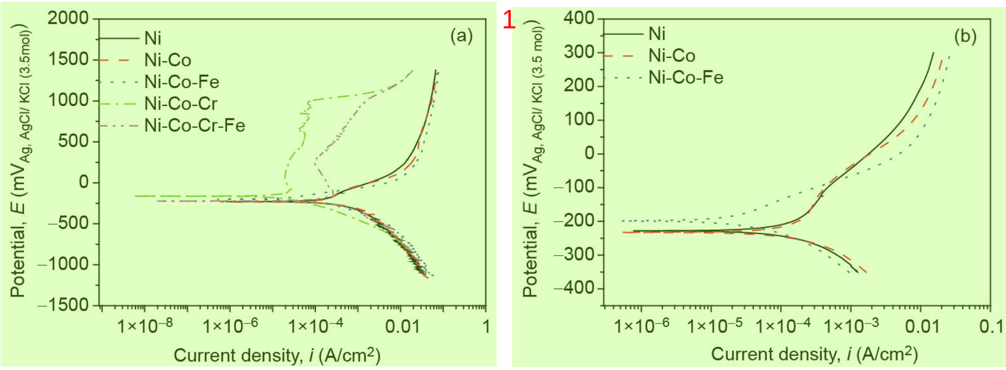

Figure 3 (Color online) Potentiodynamic polarization curves of alloys in 0.1 M HCl. (a) All the five materials; (b) Ni, NiCo and NiCoFe at the potential range of −350–300 mV.

Table 2 Electrochemical parameters of the alloys in 0.1 M HCl

Designation	E_{corr} (mV_{RE})	i_{corr} ($\mu A/cm^2$)	E_{pp} (mV_{RE})	E_b (mV_{RE})	ΔE (mV)
Ni	−228	336	–	–	–
NiCo	−233	287	–	–	–
NiCoFe	−198	228	–	–	–
NiCoCr	−163	23	−97	971	1068
NiCoCrFe	−222	208	−115	246	361

the passivation breakdown. The decrease of current density shows that the passive film is more stable and complete in the potential below 250 mV_{RE}.

Figure 4 shows the corresponding corroded surfaces of the alloys in 0.1 M HCl solution. Similarly, NiCoFe corroded worst with gully-like pits, while the pits on pure Ni and NiCo are dispersedly distributed. For pure Ni, the matrix is rough and many pits evolve in depth, indicating that the corrosion type is pitting corrosion. For NiCoCr and NiCoCrFe, only small amount of pits are shown on the corroded surfaces even in the magnification images, similar to that in the solution of NaCl. The main corrosion type for NiCoCr and NiCoCrFe is general corrosion. Differently, there are more corrosion pits on the surface of NiCoCrFe than that of NiCoCr under high magnification.

The above results indicate that the high entropy by increasing the alloying number of element is irrelevant to the high corrosion resistance in these single solid solution alloys. We need to find out the intrinsic reason for the high corrosion resistance of HEAs.

Firstly, the mechanism of the corrosion resistance of pure metals should be considered. Tuna et al. [23], Kocijan et al. [24], Sekine and Chinda [25] measured the potentiodynamic polarization curves of pure Ni, Co, Cr, Fe, and showed that the corrosion resistance order is Cr > Ni > Co > Fe. For pure Ni, the critical potential is 0.28 V higher than that of the normal hydrogen electrode (NHE) in 0.1 M NaCl at 25°C, which is almost consistent with the result measured in this test (0.345 V_{NHE}). Ni exists in the form of $NiCl_2$ in NaCl aqueous solution [26]. The oxide of Ni on the surface is NiO, which is a kind of porous structures [27]. And NiO can not protect the matrix when chloride ion erodes at higher potential.

Secondly, the alloying effects are considered. The corrosion resistance of alloys is occasionally better than the pure metals because of the synergistic function [23] and barrier function [28]. As one component corroded, the oxide of the element with better corrosion resistance will cover the total surface to prevent further corrosion. Generally, in neutral and alkaline solutions the oxide film is stable. However, in the presence of chloride ions pitting corrosion takes place, and the order of defectiveness is FeO > CoO > NiO > Cr_2O_3 [29] in the investigated alloys. In the experiment, the corrosion resistance of NiCo is only a little better than pure Ni from the potentiodynamic polarization curves of Ni and NiCo in Figure 1. It seems that there is no synergistic function in the corrosion resistance. When Fe was added to NiCo alloy, the corrosion resistance decreased badly. The oxide/hydroxide of Fe is more porous [27] and less stable than the oxide/hydroxide of Ni and Co. Fe is harmful to the corrosion resistance of alloys. Combined with Table 3, it can be notice that Fe accumulated in pits and thus the corrosion appears in that region. Table 4 shows the EDS analysis of NiCo and NiCoFe in 0.1 M HCl solution. Similar to the condition in 0.1 M NaCl, corrosion began in the region of abundant Ni for NiCo. For NiCoFe, the Fe rich region corroded first.

The addition of Cr to NiCo and NiCoFe extremely enhanced the corrosion resistance. On the surface of alloy NiCoCr, Cr might exist in the form of massive Cr_2O_3 [24,30,31], which

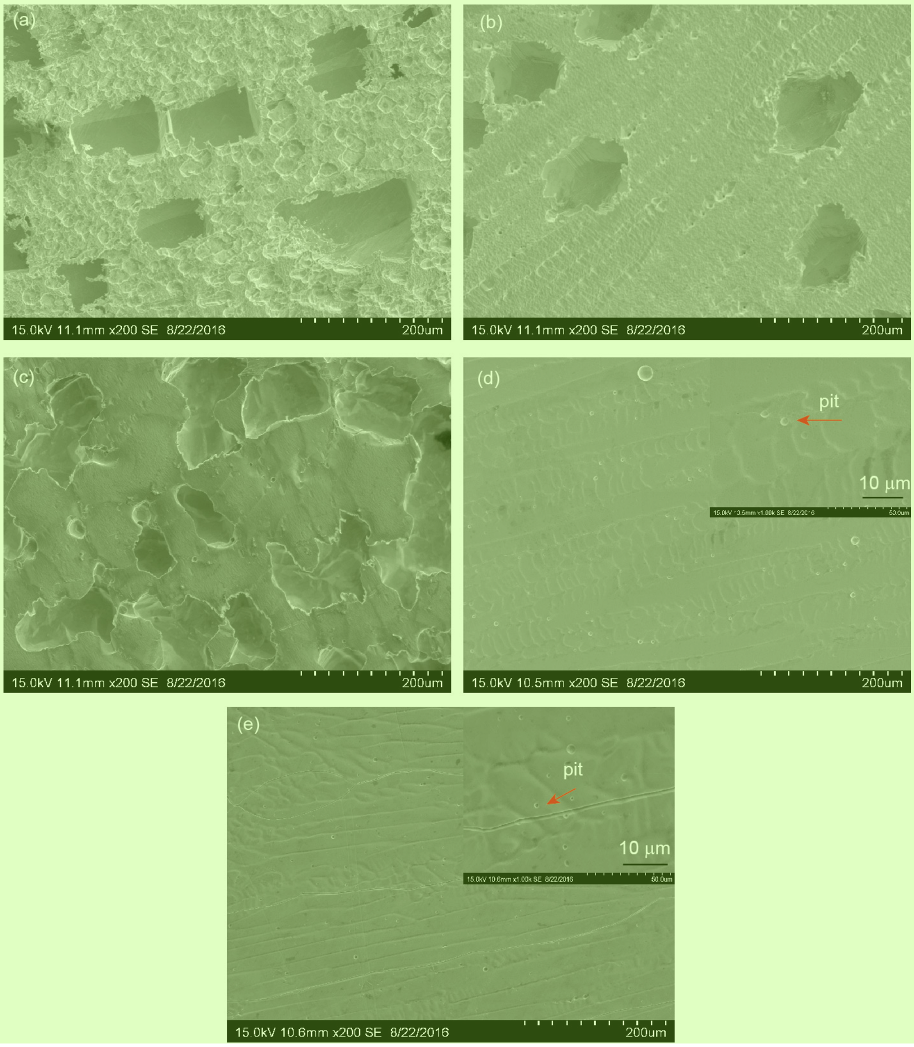

Figure 4 (Color online) The corroded surfaces of alloys in 0.1 M HCl. (a) Ni (b) NiCo; (c) NiCoFe; (d) NiCoCr; (e), NiCoCrFe.

Table 3 EDS analysis of NiCo and NiCoFe in 0.1 M NaCl

Element	NiCo (at.%)		NiCoFe (at.%)	
	Matrix	Pits	Matrix	Pits
Ni	51.15	56.37	33.45	28.64
Co	48.85	43.63	32.66	35.66
Fe	–	–	33.88	35.70

Table 4 EDS analysis of NiCo and NiCoFe in 0.1 M HCl

Element	NiCo (at.%)		NiCoFe (at.%)	
	Matrix	Pits	Matrix	Pits
Ni	50.59	53.81	33.52	11.89
Co	49.41	46.19	33.35	47.74
Fe	–	–	33.12	40.37

was highly stable in neutral solution and in acid solution, because the electrochemical behavior of Cr is immune to pH [32]. Furthermore, the existence of Cr_2O_3 was demonstrated in our test. Figure 5 shows the XPS survey spectra of NiCoCr and NiCoCrFe in 0.1 M NaCl, and the spectra contain all the alloying elements. Figure 6 plots the spectra of Cr for NiCoCr and NiCoCrFe, indicating that Cr exists mainly in the form of Cr_2O_3 [33,34]. Additionally, the XPS results show that Cr is the main element on the surface of NiCoCr and NiCoCrFe, as shown in Table 5. The location of dissolved Co and Ni on the surface would be covered by the oxide/hydroxide of Cr [24] to prevent further corrosion. What's more, the synergistic function of Ni and Cr has been widely known in stainless steel [27,35,36], including ferritic stainless steel, austenitic stainless steel and duplex stainless steel. Comparing the electrochemical behaviors of NiCo, NiCoCr, the addition of Cr is extremely helpful to corrosion resistance. It also should be noted that the content of Cr is 33.3 at.% in NiCoCr and 25 at.% in NiCoCrFe. Thus, the better corrosion resistance of NiCoCr than NiCoCrFe is due to combined function of higher Cr content and absence of detrimental Fe.

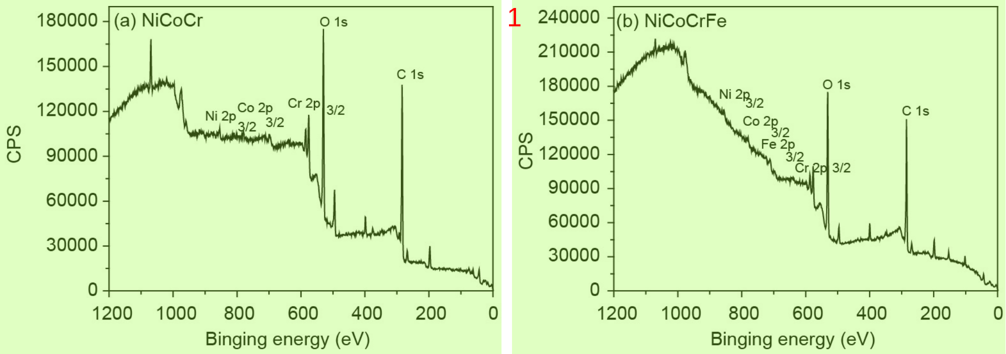

Figure 5 The XPS survey spectra of NiCoCr and NiCoCrFe in 0.1 M NaCl. (a) NiCoCr; (b) NiCoCrFe.

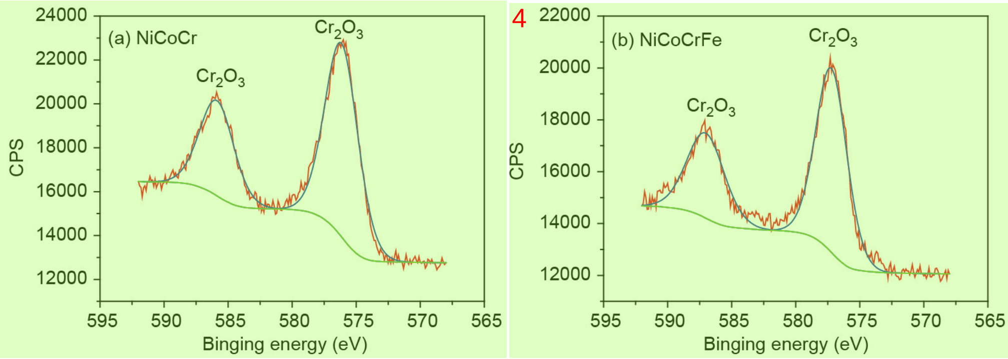

Figure 6 (Color online) The XPS survey spectra of Cr for NiCoCr and NiCoCrFe in 0.1 M NaCl. (a) NiCoCr; (b) NiCoCrFe.

Table 5 The XPS result of NiCoCr and NiCoCrFe in 0.1 M NaCl

Element	NiCoCr (at.%)	NiCoCrFe (at.%)
Ni	8.77	9.59
Co	5.85	12.53
Cr	85.38	59.59
Fe	–	18.29

Based on the above analyses, we can simply divide the investigated alloys into two classes, one contains Cr, NiCoCr and NiCoCrFe, and the other is without Cr. Remarkably, the group containing Cr is far better than the alloys without Cr with higher breakdown potential (E_b), higher corrosion potential (E_{corr}) and lower current density. Accordingly, the element of Cr within the solid solution plays the dominant role in the equal atomic alloys (or HEAs) instead of the simple phase composition.

4 Conclusions

The electrochemical properties and corroded surfaces of pure Ni, equal atomic NiCo, NiCoFe, NiCoCr and NiCoFeCr were investigated. The results clearly show the decisive role of solid soluted Cr in the corrosion resistance of HEAs, because of the formation of Cr_2O_3, which is highly stable and compact. The corrosion resistance of FCC high entropy alloys is mainly dependent on the effect of Cr, and nothing to do with the entropy of mixing. This discovery may inspire us to design corrosion resistant HEAs more reasonably and efficiently.

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